INCH-POUND

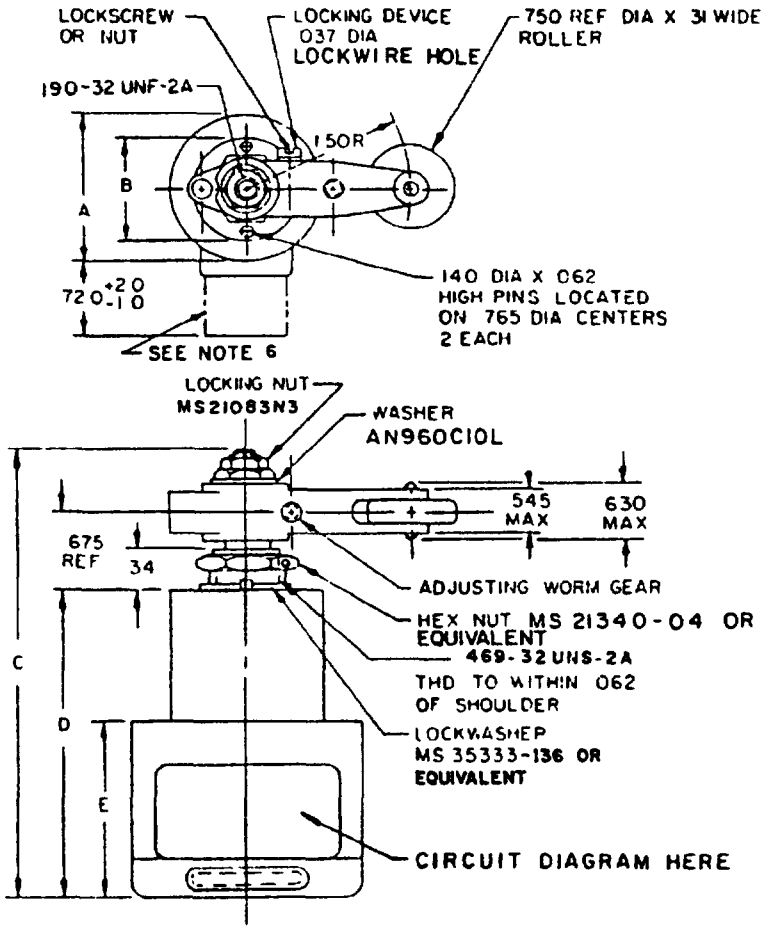
MIL-S-8805/48F
3 December 1990
SUPERSEDING
MIL-S-8805/48E
17 March 1983

MILITARY SPECIFICATION SHEET

SWITCHES, SENSITIVE, LIMIT, ROLLER LEVER, SELF-RETURN, RESILIENT SEAL, FLUID RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-8805.

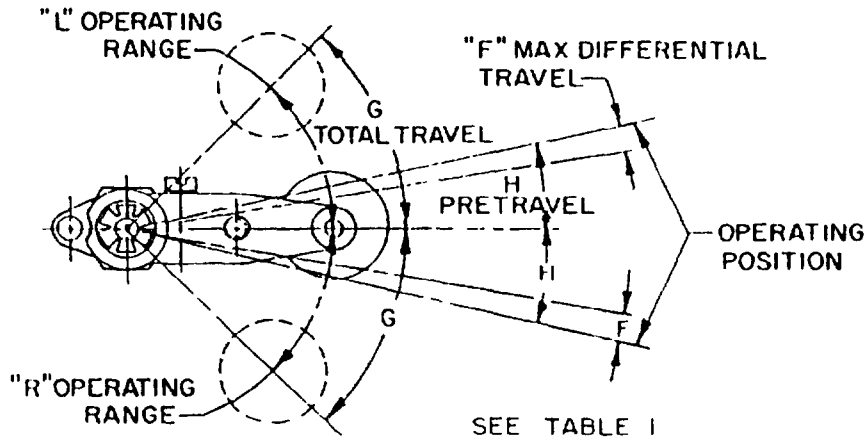


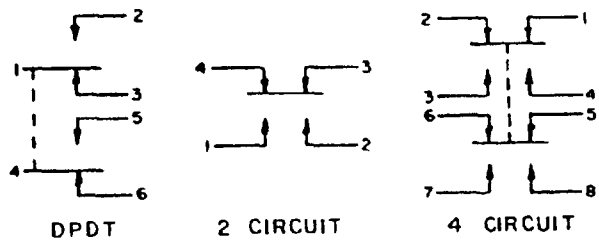
Inches	mm
.037	0.94
.062	1.57
.140	3.56
.190	4.83
.31	7.8
.34	8.6
.469	11.91
.545	13.84
.630	16.00
.675	17.15
.750	19.05
.765	19.43
1.0	25.
1.50	38.1
2.0	50.
72.0	1828.

FIGURE 1. Configuration and dimensions.

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CIRCUIT SCHEMATICS

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Millimeters are in parentheses.
4. Unless otherwise specified, tolerances are ±.02 (0.51 mm) for two place decimals and ±.005 (0.13 mm) for three place decimals.
5. Envelope optional provided dimensions specified are not exceeded.
6. Leads shall be marked in accordance with MIL-W-5088 with the switch circuit identification number followed by wire gauge number (1-20, 2-20, etc.). Lead wire shall be MIL-W-22759/7.
7. Lever shall be adjustable for an infinite number of positions through 360° without removing lever.

FIGURE 1. Configuration and dimensions - Continued.

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REQUIREMENTS.

Dimensions and configuration See figure 1 and table I.

TABLE I. Dimensions and operating characteristics.

MS21320-	Operating range	A diameter maximum	B diameter maximum	C maximum	D maximum	E maximum	F maximum	G ±3°	H	Lead wires: Size	Lead wires: Number required
1	R	1.030 (26.16)	1.030 (26.16)	3.41 (86.61)	2.16 (54.86)	---	4°	45°	13° ±3°	Number 20	6
2	L										
3	R	1.530 (38.86)		4.10 (104.14)	2.82 (71.63)	1.86 (47.24)	10°		20° ±4°	Number 18	4
4	L										
5	R			3.68 (93.47)	2.44 (61.98)	1.46 (37.08)	5°		15° ±4°		
6	L										
7	R			4.10 (104.14)	2.82 (71.63)	1.86 (47.24)	10°		15° ±5°		8
8	L										

Enclosure design: Four (resilient). All entrances to the switch cavity except through the actuator bushing shall be sealed by fusion of glass-to-metal, metal-to-metal, or ceramic-to-metal and the lead wires shall be potted to provide stress relief.

Temperature characteristics: One (-55°C to +85°C).

Shock type: M (100 G).

Sinusoidal vibration grade: One (10 to 500 Hz).

Material:

Roller: Plastic or laminated phenolic.

Plastic:

MS21320-1 and -2: Type CFG in accordance with MIL-M-14 may be used internally.

MS21320-3 and -4: General purpose mineral and flock-filled phenolic may be used internally.

Finish. Switch housing shall be processed to resist corrosion.

Maximum weight with leads:

MS21320-1, -2: 9.5 ounces.

MS21320-3, -4: 15.5 ounces.

MS21320-5, -6: 9.5 ounces.

MS21320-7, -8: 15.5 ounces.

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Operating characteristics

Operating torque. 12 to 25 inch-pounds.

Release torque

MS21320-1, -2: 9 inch-pounds minimum.
MS21320-3, -4: 4 inch-pounds minimum.
MS21320-5, -6, -7, -8. 8 inch-pounds minimum.

Full overtravel torque. 40 inch-pounds maximum.

} ±10 percent variation from specified values acceptable after test.

Coincidence of operating and releasing points: Not applicable.

Contact resistance. Not applicable.

Insulation resistance 100 meghoms minimum.

Dielectric withstanding voltage.

At atmospheric pressure: 1,000 V rms.

At reduced barometric pressure: 50,000 feet; 400 V rms.

Mechanical endurance: 25,000 cycles.

Electrical endurance. 25,000 cycles.

Electrical ratings: See table II.

Fluid resistance: Except for the cut end of the lead wire, switches shall be submerged in each of the following fluids for 2 minutes to 2 minutes 30 seconds, which shall consist of one cycle (one cycle is 10 minutes to 12 minutes 30 seconds total). Each switch shall be subjected to three cycles.

a. Turbine fuel (MIL-T-5624).

b. Hydraulic fluid (SAE AS1241A).

c. Coolanol 1/ (MIL-C-47220).

d. Ethylene glycol (MIL-E-9500).

e. Lubricating oil (MIL-L-7808).

After each immersion, the excess fluid shall be blown off the external surfaces of the switch with an air jet. Following the third cycle, the switch shall be subjected to and shall meet the requirements for dielectric withstanding voltage, insulation resistance, operating characteristics, seal tests, and marking visibility.

Marking: The circuit schematic shall be marked on the switch case.

Part or Identifying Number (PIN): See table III.

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TABLE II Electrical ratings

Rating code	Load				
	Sea level 28 V dc			50,000 ft 28 V dc	
	Resistive	Inductive	Motor	Resistive	Inductive
	(amperes)	(amperes)	(amperes)	(amperes)	(amperes)
A	5	3	4	5	2
B	10	6	6	10	3
C	15	10	5	15	10

TABLE III. PIN and characteristics.

PIN	Electrical rating code (table II)	Circuit
MS21320-1	A	DPDT
MS21320-2	A	DPDT
MS21320-3	B	DPDT
MS21320-4	B	DPDT
MS21320-5	C	2 circuit
MS21320-6	C	2 circuit
MS21320-7	C	4 circuit
MS21320-8	C	4 circuit

Qualification inspection:

Group submission: See table IV.

Group A inspection:

Seal test: Only watertight test shall be performed.

TABLE IV. Qualification inspection (group submission).

Examination or test	Samples	Extent of approval
Qualification inspection table of MIL-S-8805	MS21320-7 or -8 (24 units)	All
Visual and mechanical examination, group II	MS21320-1, -3 or MS21320-2, -4 (2 units each)	
Group VII	MS21320-1, -3 or MS21320-2, -4 (10 units each)	

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Revision letters are not used to denote changes due to the extensiveness of the changes

CONCLUDING MATERIAL

Custodians
Army - ER
Navy - EC
Air Force - 85

Review activities
Army - AR
Navy - AS, OS
Air Force - 99
DLA - ES

User activities:
Army - AV, MI
Navy - MC
Air Force - 19

Preparing activity:
Navy - EC

Agent
DLA - ES

(Project 5930-1413-03)

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